

FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

ABSTRACT OF DOCTORAL THESIS

Mgr. Petra Barančíková

Evaluating Metrics for Machine Translation Evaluation: From Paraphrases to Semantic Embeddings

Institute of Formal and Applied Linguistics

Supervisor:	doc. RNDr. Markéta Lopatková, Ph.D.
Consultant:	prof. RNDr. Ondřej Bojar, Ph.D.
Study programme:	Computational Linguistics
Study branch:	Mathematical Linguistics

Prague 2026

The results of this thesis were achieved in the period of doctoral study at the Faculty of Mathematics and Physics, Charles University.

Student:	Mgr. Petra Barančíková
Supervisor:	doc. RNDr. Markéta Lopatková, Ph.D. Institute of Formal and Applied Linguistics Faculty of Mathematics and Physics Charles University Malostranské nám. 25, 118 00 Praha 1
Consultant:	prof. RNDr. Ondřej Bojar, Ph.D. Institute of Formal and Applied Linguistics Faculty of Mathematics and Physics Charles University Malostranské nám. 25, 118 00 Praha 1
Department:	Institute of Formal and Applied Linguistics Faculty of Mathematics and Physics Charles University Malostranské nám. 25, 118 00 Praha 1
Opponents:	doc. Mgr. Pavel Rychlý, Ph.D. Masarykova univerzita
	doc. Jernej Vičič, Ph.D. University of Primorska

The thesis defence will take place on June 24, 2026 before the committee for thesis defences in Mathematical Linguistics at the Faculty of Mathematics and Physics, Charles University, Malostranské nám. 25, 118 00 Praha 1, room Malá aula.

The thesis can be viewed at the Study Department of Doctoral Studies of the Faculty of Mathematics and Physics, Charles University, Ke Karlovu 3, 121 16 Praha 2.



MATEMATICKO-FYZIKÁLNÍ
FAKULTA
Univerzita Karlova

AUTOREFERÁT DISERTAČNÍ PRÁCE

Mgr. Petra Barančíková

Hodnocení metrik pro hodnocení strojového překladu: od parafrází k sémantickým embeddingům

Ústav formální a aplikované lingvistiky

Školitelka:	doc. RNDr. Markéta Lopatková, Ph.D.
Konzultant:	prof. RNDr. Ondřej Bojar, Ph.D.
Studijní program:	Počítačová lingvistika
Studijní obor:	Matematická lingvistika

Praha 2026

Výsledky uvedené v této disertační práci byly získány během doktorského studia na Matematicko-fyzikální fakulta, Univerzita Karlova.

Doktorandka:	Mgr. Petra Barančíková
Školitelka:	doc. RNDr. Markéta Lopatková, Ph.D. Ústav formální a aplikované lingvistiky Matematicko-fyzikální fakulta Univerzita Karlova Malostranské nám. 25, 118 00 Praha 1
Konzultant:	prof. RNDr. Ondřej Bojar, Ph.D. Ústav formální a aplikované lingvistiky Matematicko-fyzikální fakulta Univerzita Karlova Malostranské nám. 25, 118 00 Praha 1
Pracoviště:	Ústav formální a aplikované lingvistiky Matematicko-fyzikální fakulta Univerzita Karlova Malostranské nám. 25, 118 00 Praha 1
Oponenti:	doc. Mgr. Pavel Rychlý, Ph.D. Masarykova univerzita
	doc. Jernej Vičič, Ph.D. Univerza na Primorskem

Obhajoba disertační práce se koná dne 22. září 2026 před komisí pro obhajoby disertačních prací v oboru Matematická lingvistika na Matematicko-fyzikální fakulta, Univerzita Karlova, Malostranské nám. 25, 118 00 Praha 1, místnost Malá aula.

S disertační prací je možné se seznámit na oddělení doktorského studia Matematicko-fyzikální fakulta, Univerzita Karlova, Ke Karlovu 3, 121 16 Praha 2.

Contents

Introduction **1**

1 Machine Translation Evaluation **3**

2 Targeted Sentence Paraphrasing for MT Evaluation **4**

Simple substitution. 4

PARMESAN. 4

Paraphrasing via machine translation. 5

Findings. 5

3 Sentence Embedding Evaluation **6**

Linear relations in embedding spaces. 6

The COSTRA dataset. 7

Intrinsic vs. extrinsic evaluation. 7

Conclusion **9**

List of Publications **16**

Introduction

During the last decade, machine translation (MT) systems have changed significantly progressing from phrase-based statistical systems to neural models and, more recently, to large language models. Although translation quality has improved dramatically, reliable evaluation of translation outputs remains challenging. Reliable evaluation is still very important: it is needed to compare systems, guide model development, interpret shared-task results, and make research claims comparable across time.

Human evaluation remains the gold standard for assessing translation quality. However, it is expensive, time-consuming, subjective, and difficult to reproduce at scale. Consequently, automatic metrics are important component of MT research.

The main challenge for machine translation is that a single source sentence can be translated correctly in many different ways. A useful metric must decide whether a hypothesis and a reference translation, i.e. sentence translated by professional human translator, express the same meaning even when they differ lexically, morphologically, or syntactically.

Rich morphology and relatively free word order of the Czech language allow many surface variants that preserve the same meaning. Traditional string-based metrics such as BLEU can penalize correct translations simply because they differ from the reference wording (see Figure 1). Modern neural metrics reduce this limitation by relying on learned semantic representations, but they introduce new questions: what do sentence embeddings actually encode, can they reliably encode the meaning of sentence?

Source	*Banks are testing payment by mobile telephone*					
Hyp.	Banky	zkoušejí	platbu	pomocí	mobilního	telefonu
	Banks	are testing	payment	by	mobile	phone
	Banks are testing payment by mobile phone					
Ref.	Banky	testují	placení	mobilem		
	Banks	are testing	paying	by mobile phone		
	Banks are testing paying by mobile phone					

Figure 1: In this example from Workshop on Machine Translation (WMT) 2012 [Callison-Burch et al., 2012], despite the hypothesis being grammatically correct and having the same meaning as the source sentence, there is only a single overlapping unigram between the hypothesis and the reference translation. As a result, the hypothesis would receive a very low score when evaluated using n-gram overlap-based automatic evaluation metrics.

This dissertation follows both sides of this development. The first part revisits the traditional setting and asks whether string-based evaluation can be improved by automatically paraphrasing reference translations. The second part studies sentence embeddings as a modern semantic alternative, examining their geometric structure, their behavior on Czech semantic transformations, and their performance inside neural MT evaluation metrics.

Although the underlying technologies differ substantially, both parts address the same research question: how automatic evaluation can better capture semantic equivalence between translations. More specifically, the thesis addresses four research questions:

- Can the reliability of traditional string based methods be improved through targeted paraphrasing of reference sentences?
- Do sentence embeddings exhibit systematic geometric structure corresponding to semantic transformations?
- How well do modern sentence embedding models capture complex semantic transformations in Czech?
- Does strong perfomance on intrinsic semantic benchmarks imply strong performance as a machine translation evaluation metric?

Chapter 1

Machine Translation Evaluation

The thesis first establishes the evaluation setting in which the experiments are interpreted. We present a detailed survey of machine translation evaluation methods from the last twenty years at the Workshop on Machine Translation (WMT) shared tasks from adequacy and fluency judgments through relative ranking to Direct Assessment, MQM, and Error Span Annotation. These protocols differ in cost, granularity, and reproducibility, but they share the same goal: to estimate how much a translation preserves the meaning of the source. They are also important because their scores serve as training or validation targets for many automatic metrics.

Automatic evaluation metrics attempt to approximate human judgments efficiently and reproducibly. Traditional metrics such as BLEU [Papineni et al., 2002], TER, and character-level metrics measure surface overlap between a system hypothesis and one or more human references. Their advantages are practical: they are fast, deterministic, transparent, and comparable with a large body of previous work. Their weakness is conceptual: lexical overlap is only an imperfect proxy for translation quality. Correct translations may use different words, inflectional forms, or word order than the reference.

This limitation motivated metrics that use linguistic resources. Metrics such as Meteor and Meteor-NEXT add stemming, synonym matching, and paraphrase tables, making the comparison more tolerant of legitimate variation. However, the added resources also introduce noise and language dependence. In the Czech setting, automatically created paraphrase tables contain many pairs that are not valid paraphrases in context. When these pairs are used directly inside a metric, the metric can reward grammatical errors, inappropriate substitutions, or partially untranslated output.

Recent neural metrics, including BLEURT [Sellam et al., 2020] and COMET [Rei et al., 2020], shift the problem from explicit string matching to learned semantic representations. They often correlate better with human judgments, but they are less transparent and depend on training data, model architecture, and evaluation protocol. At the same time, traditional metrics remain widely reported in MT research [Marie et al., 2021]. For this reason, the thesis studies both paradigms: explicit paraphrasing for string-based metrics and sentence embeddings for neural semantic evaluation.

Chapter 2

Targeted Sentence Paraphrasing for MT Evaluation

The first experimental part investigates targeted paraphrasing of reference translations. The motivation is simple: if a metric penalizes a valid hypothesis because it is lexically different from the reference, the evaluation can be improved by generating additional reference variants that are closer to the hypothesis while preserving the original meaning. This approach keeps the metric itself mostly unchanged and instead improves the material against which the hypothesis is compared.

Simple substitution. The first method is deliberately conservative. It identifies candidate paraphrase pairs between a hypothesis and its reference, validates them against lexical and phrasal paraphrase resources, substitutes selected expressions into the reference, and optionally applies Depfix to repair grammatical errors introduced by the substitution. The method is inspired by earlier work on paraphrasing for automatic evaluation [Kauchak and Barzilay, 2006], but it is adapted to Czech morphology and to the limited reliability of Czech paraphrase resources.

This work demonstrated that evaluation quality can be improved without modifying BLEU itself, simply by enriching the reference space. The strongest improvements come from one-word substitutions selected from higher-quality resources and followed by grammatical correction. Multi-word paraphrases increase coverage, but they are more vulnerable to noise. The main lesson is that paraphrase precision and grammatical well-formedness matter more than the complexity of the generation method.

Parmesan. The next experiment asks whether paraphrased references are still useful for metrics that already include paraphrase matching internally, especially METEOR-NEXT. Several variants of METEOR-NEXT are compared with progressively filtered paraphrase tables and with three reference types: original references, paraphrased references, and Depfix-corrected paraphrased references. The best configuration removes internal paraphrase matching from METEOR-NEXT and evaluates Depfix-corrected paraphrased references using exact matching.

This leads to the PARMESAN metric, which shows that paraphrases are more effective as alternative references than as internal matching rules. The result is conceptually important: paraphrases help when they become grammatical alternative references, but noisy paraphrase tables can harm evaluation when they are used directly as metric-internal matching rules. The method therefore separates two tasks that are often conflated: generating plausible reference variants and scoring the final hypothesis-reference match.

Paraphrasing via machine translation. The thesis also explores whether more advanced methods of paraphrasing can improve on the simple substitution method. Paraphrasing and translation are closely related tasks, they both aim to preserve meaning of a sentence while changing its surface form, however there are much more resources and tools available for translation than for paraphrasing. Therefore, we adapted existing machine translation systems for monolingual paraphrasing.

First, we adapt Moses, a phrase-based statistical MT system, for monolingual Czech-to-Czech paraphrasing. Czech paraphrase tables are used as translation tables, and a new decoder feature guides generation toward paraphrases that overlap with the evaluated hypothesis. Further, we experiment with adapting Treex, a deep-syntactic transfer-based MT system, for monolingual targeted paraphrasing.

These systems are more expressive, but they do not consistently outperform the simpler method. Their outputs can introduce broader structural changes and additional errors, making the resulting references less controlled and sometimes less useful for evaluation. This finding is one of the main empirical results of the first part: for MT evaluation, a simple, precise, grammatically corrected paraphrase can be more valuable than a sophisticated generator.

Findings. Overall, the paraphrasing experiments show that reference expansion can improve traditional metrics for Czech MT evaluation, but only when paraphrases are targeted, precise, and grammatically sound. Larger paraphrase resources are not automatically better because noise can outweigh coverage. The results remain relevant even in the era of neural metrics: string-based metrics are still widely reported, and the paraphrasing experiments make explicit where surface matching fails.



Figure 3.1: An illustration reprinted from Barančíková and Bojar [2019]. The upper plot represents a continuous multi-dimensional vector space where individual sentences are represented as points (the "space of sentences"). Pairs of related sentences are connected by arrows, with different dashing styles indicating various relationship types.
The lower plot illustrates the corresponding "space of operations." Here, a vector difference is shown, with all arrows moved to start at a common origin.
The underlying hypothesis is that consistent semantic changes (e.g., all vector transformations extracted from sentence pairs differing in the speed of travel "running instead of walking") should be represented close to each other in the space of operations, i.e., form a more or less compact *cluster*.

Chapter 3

Sentence Embedding Evaluation

In the second part, we focus on sentence embeddings evaluation. Embedding-based metrics represent sentences as points in a continuous vector space, where geometric similarity is expected to correspond to semantic similarity. This decreases the need for explicit paraphrasing, but it introduces a new question: what semantic information is actually encoded in sentence embeddings, and is it encoded in a way that is useful for MT evaluation?

Linear relations in embedding spaces. In the first section of the second part, we investigate whether sentence embeddings exhibit consistent linear structures analogous to word embeddings. As Mikolov et al. [2013] demonstrated, word embeddings capture semantic and syntactic relationships geometrically, which they evaluated through tests like adjective declension (easy" → easier" → easiest"), verb tense changes (walking" → walk"), and capital-country matching (Athens" → "Greece").

Our objective is to determine whether *whole sentence* embeddings exhibit the same linear behavior, looking for an analogous behavior to that observed with word

embeddings. This idea is illustrated in Figure 3.1. Specifically, we examine whether semantic shifts between sentences can be modeled as transformation vectors. Understanding whether sentence embeddings exhibit consistent linear structure is particularly important for embedding-based evaluation metrics, which implicitly rely on geometric properties such as vector similarity and compositionality.

The results show that some sentence embedding models do contain coherent geometric structure. Certain lexical and semantic changes form recognizable clusters, indicating that the models encode more than arbitrary surface similarity. At the same time, this structure depends strongly on the training objective. Architecture alone is not sufficient: untrained or poorly suited models do not show the same behavior. Negation and more complex logical changes remain difficult across models.

The Costra dataset. The major contribution of this thesis is Costra, a Czech benchmark for evaluating sentence embeddings [Barančíková and Bojar, 2020a,b]. It consists of 7 thousand manually created sentences and 16 thousand pairwise human similarity judgments for evaluation. The dataset was designed to test whether sentence embeddings capture complex semantic relations beyond surface similarity. The sentences are selected to create a real challenge for the embeddings, e.g., what is closer to original sentence - a distinct paraphrase or sentence that is similar on the surface but different in meaning? Other categories include temporal changes, modality, opposition, generalization or style changes.

Experiments on Costra show that current sentence encoders still rely heavily on surface similarity. Even strong models struggle when lexical overlap is high but meaning changes substantially, and they often fail to capture ordinal semantic progressions. Czech-specific and multilingual models each have strengths, but no tested model fully solves the diagnostic tasks. The Costra results therefore reveal limitations that are often hidden by standard semantic textual similarity benchmarks.

Intrinsic vs. extrinsic evaluation. In the previous section, we demonstrated through intrinsic evaluation that sentence embeddings often rely on surface-level similarities rather than capturing complex linguistic phenomena. However, the primary focus of this thesis is extrinsic evaluation — machine translation evaluation (MTE).

In contrast to the simple mathematical computations used in intrinsic evaluation, extrinsic evaluation for MTE requires costly model training. This leads to a natural question: can intrinsic evaluation serve as a predictor for downstream performance? Semantic Textual Similarity (STS) is closely related to MTE, as both tasks assess whether two sentences convey the same meaning despite surface differences. If intrinsic benchmarks correlated well with extrinsic performance, they could be used as a quick proxy for selecting promising embedding models.

This question is particularly relevant given the growing number of available embedding models. Multilingual encoders such as and provide strong cross-lingual representations, while language-specific models aim to better capture properties of individual languages. For morphologically rich languages such as Czech, it is not

obvious which approach is more suitable for MT evaluation. This raises another question: do Czech-specific encoders outperform multilingual models in evaluation of Czech translations?

In the final experiments, we compare intrinsic evaluation of Czech sentence embeddings with their downstream usefulness in MT evaluation [Barančíková and Bojar, 2025]. Several Czech and multilingual encoders are evaluated on Costra and Czech STS datasets. The same encoders are then used inside COMET-based metrics trained for machine translation evaluation and quality estimation on WMT English-to-Czech data.

The results reveal a striking mismatch. Models that perform best on intrinsic semantic benchmarks do not necessarily become the best MT evaluation metrics after fine-tuning. Conversely, some models with weak intrinsic performance become highly competitive inside COMET-based metrics. Task-specific fine-tuning changes the model ranking substantially, which challenges the assumption that intrinsic semantic quality is a reliable shortcut for choosing encoders for MT evaluation.

The comparison also shows that reference-free quality estimation can become competitive with reference-based evaluation in this setup. This agrees with recent developments in MT evaluation, where source-aware and reference-free neural metrics often perform on par with reference-based metrics. The thesis therefore concludes that embeddings must be selected and validated with respect to the target task, not only with respect to standalone semantic benchmarks.

Conclusion

This thesis examines two ways of addressing the same fundamental problem: automatic metrics must evaluate meaning, but meaning is not tied to a single surface form. The first part uses explicit paraphrases to make traditional metrics more tolerant of valid translation variants. The second part analyzes sentence embeddings as implicit semantic representations used by modern neural metrics.

Across both parts, the same conclusion emerges: the most complex method is not necessarily the most effective one. Simple lexical paraphrases, when selected carefully and corrected grammatically, improve evaluation more reliably than more ambitious paraphrase generators. Similarly, larger or intrinsically stronger embedding models do not automatically produce better MT evaluation metrics after fine-tuning. What matters is alignment between the method, the data, and the specific notion of semantic equivalence required by the evaluation task.

The dissertation contributes practical resources and methodological insights: targeted paraphrasing methods for Czech MT evaluation, the PARMESAN metric, an analysis of linear structure in sentence embeddings, the COSTRA benchmark for Czech sentence embeddings, and an empirical comparison of intrinsic and extrinsic embedding evaluation. Together, these contributions document the transition from surface-based evaluation toward semantic evaluation while showing that semantic evaluation remains difficult, language-sensitive, and task-dependent.

This thesis explores two complementary approaches to automatic machine translation evaluation to address the challenges of measuring semantic equivalence: targeted paraphrasing for traditional string-based metrics and the analysis of modern sentence embeddings.

—- This thesis examines two complementary approaches to automatic machine translation evaluation. The first part focuses on targeted paraphrasing of reference translations as a method to improve the reliability of traditional string-based evaluation metrics. The second part examines the behavior and limitations of modern sentence embedding models with respect to machine translation evaluation. Together, these two approaches highlight the difficulty of measuring semantic equivalence in machine translation.

In addition to the experimental work, the thesis provides a structured overview of machine translation evaluation methods, with a particular focus on human evaluation in the WMT shared tasks. We trace the development from early adequacy and fluency judgments through relative ranking to Direct Assessment, MQM, and more recent approaches such as ESA. This overview clarifies what current evaluation protocols actually measure, as well as their practical limitations, including subjectivity, cost, and limited reproducibility. It also provides an

essential context for interpreting automatic metrics, which are typically trained to approximate these human judgments.

Building upon this theoretical foundation, the main contributions of this thesis can be summarized as follows. First, we demonstrate that the reliability of string-based metrics such as BLEU can be systematically improved — specifically in the context of a morphologically rich language like Czech — by creating artificial reference sentences closer to machine translation outputs. We show that for Czech, simple substitution-based paraphrasing methods consistently outperform more complex approaches based on statistical or rule-based machine translation systems. This finding suggests that controlled lexical variation is more effective than generating complete paraphrases that differ more significantly in structure. At the same time, we show that large, automatically created paraphrase resources can reduce evaluation quality by introducing noise. Filtering and moving paraphrase processing from evaluation to preprocessing improves results. Finally, we establish that grammatical correctness plays a crucial role: linguistically well-formed paraphrases consistently improve evaluation reliability.

Second, we provide a detailed analysis of sentence embeddings as a semantic evaluation tool. We analyze the geometric properties of both English and multilingual sentence embedding spaces and show that certain models exhibit a clear and interpretable geometric structure, in which simple linguistic transformations correspond to consistent directional shifts. This structure is not a trivial byproduct of architecture, but emerges from the training objective, particularly in models trained with semantic or contrastive signals.

To support further examination of embedding properties, we introduce the COSTRA dataset of complex Czech sentence transformations, addressing a gap in existing evaluation resources. Our experiments show that even strong, state-of-the-art embedding models continue to rely heavily on surface-level similarity and often fail to capture more complex semantic relations. We also find that strong performance on intrinsic evaluation tasks does not reliably predict performance in machine translation evaluation, despite both being designed to assess semantic similarity.

Taken together, the results lead to a consistent conclusion: a more complex model does not automatically result in better evaluation performance. Simple string-based metrics, when supported by carefully constructed and grammatically sound paraphrases, can perform as well as or better than more complex approaches. Similarly, larger and more general sentence embedding models do not necessarily outperform smaller or more specialized models when applied to machine translation evaluation. These findings demonstrate that evaluation quality depends primarily on how well a method is matched to the specific task, rather than on model complexity alone.

At the same time, several open questions remain. In particular, it is unclear why intrinsic and extrinsic evaluation lead to contradictory conclusions about embedding quality. A deeper understanding of this mismatch would require a more precise characterization of what aspects of meaning are captured by human evaluation and how these relate to current intrinsic benchmarks. We leave this as future work: designing intrinsic evaluation methods that better reflect the requirements of downstream tasks such as machine translation evaluation.

Another promising direction for future work is the deeper exploration of sentence representations with interpretable geometric properties, as observed in our analysis of embedding spaces. Understanding what training conditions lead to these structured spaces—and how to mitigate their failures in areas like logical negation—may contribute to better generalization and more interpretable models. —

Bibliography

Petra Barančíková and Ondřej Bojar. COSTRA 1.0: A Dataset of Complex Sentence Transformations. In Nicoletta Calzolari, Frédéric Béchet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis, editors, *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3535–3541, Marseille, France, May 2020a. European Language Resources Association. ISBN 979-10-95546-34-4. URL https://aclanthology.org/2020.lrec-1.434/.

Petra Barančíková and Ondřej Bojar. Costra 1.1: An Inquiry into Geometric Properties of Sentence Spaces. In *23rd International Conference on Text, Speech and Dialogue*, pages 135–143, 2020b.

Petra Barančíková and Ondřej Bojar. Intrinsic vs. Extrinsic Evaluation of Czech Sentence Embeddings: Semantic Relevance Doesn't Help with MT Evaluation. In Pierrette Bouillon, Johanna Gerlach, Sabrina Girletti, Lise Volkart, Raphael Rubino, Rico Sennrich, Ana C. Farinha, Marco Gaido, Joke Daems, Dorothy Kenny, Helena Moniz, and Sara Szoc, editors, *Proceedings of Machine Translation Summit XX: Volume 1*, pages 265–275, Geneva, Switzerland, June 2025. European Association for Machine Translation. ISBN 978-2-9701897-0-1. URL https://aclanthology.org/2025.mtsummit-1.20/.

Chris Callison-Burch, Philipp Koehn, Christof Monz, Matt Post, Radu Soricut, and Lucia Specia. Findings of the 2012 Workshop on Statistical Machine Translation. In *Seventh Workshop on Statistical Machine Translation*, pages 10–51, Montréal, Canada, 2012.

David Kauchak and Regina Barzilay. Paraphrasing for Automatic Evaluation. In *Proceedings of the main conference on Human Language Technology Conference of the North American Chapter of the Association of Computational Linguistics*, HLT-NAACL '06, pages 455–462, Stroudsburg, PA, USA, 2006. Association for Computational Linguistics. doi: 10.3115/1220835.1220893.

Benjamin Marie, Atsushi Fujita, and Raphael Rubino. Scientific Credibility of Machine Translation Research: A Meta-Evaluation of 769 Papers. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7297–7306, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.566. URL https://aclanthology.org/2021.acl-long.566.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient Estimation of Word Representations in Vector Space. *CoRR*, abs/1301.3781, 2013. URL `http://arxiv.org/abs/1301.3781`.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a Method for Automatic Evaluation of Machine Translation. In Pierre Isabelle, Eugene Charniak, and Dekang Lin, editors, *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA, July 2002. Association for Computational Linguistics. doi: 10.3115/1073083.1073135. URL `https://aclanthology.org/P02-1040/`.

Petra Barančíková and Ondřej Bojar. In Search for Linear Relations in Sentence Embedding Spaces. In *Proceedings of the 19th Conference ITAT 2019: Slovenskočeský NLP workshop (SloNLP 2019)*, pages 125–132, Košice, Slovakia, 2019. CreateSpace Independent Publishing Platform.

Ricardo Rei, Craig Stewart, Ana C Farinha, and Alon Lavie. COMET: A Neural Framework for MT Evaluation. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.213. URL `https://aclanthology.org/2020.emnlp-main.213/`.

Thibault Sellam, Dipanjan Das, and Ankur P. Parikh. BLEURT: Learning Robust Metrics for Text Generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7881–7892. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.704. URL `https://aclanthology.org/2020.acl-main.704`.

Bibliography

Petra Barančíková and Ondřej Bojar. In search for linear relations in sentence embedding spaces. In *Proceedings of the 19th Conference ITAT 2019: Slovenskočeský NLP Workshop*, pages 125–132, Košice, Slovakia, 2019.

Petra Barančíková and Ondřej Bojar. COSTRA 1.0: A dataset of complex sentence transformations. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3535–3541, Marseille, France, 2020.

Petra Barančíková and Ondřej Bojar. COSTRA 1.1: An inquiry into geometric properties of sentence spaces. In *Proceedings of Text, Speech, and Dialogue*, pages 135–143, Cham, Switzerland, 2020.

Petra Barančíková and Ondřej Bojar. Intrinsic vs. extrinsic evaluation of Czech sentence embeddings: Semantic relevance doesn't help with MT evaluation. In *Proceedings of Machine Translation Summit XX: Volume 1*, pages 265–275, Geneva, Switzerland, 2025.

Petra Barančíková, Rudolf Rosa, and Aleš Tamchyna. Improving evaluation of English–Czech MT through paraphrasing. In *Proceedings of LREC 2014*, pages 3402–3407, Reykjavík, Iceland, 2014.

Petra Barančíková and Rudolf Rosa. Targeted paraphrasing on deep syntactic layer for MT evaluation. In *Proceedings of Depling 2015*, pages 20–27, Uppsala, Sweden, 2015.

Ondřej Bojar, Christian Buck, Chris Callison-Burch, Barry Haddow, Philipp Koehn, Christof Monz, Matt Post, Radu Soricut, and Lucia Specia. Scratching the surface of possible translations. In *Proceedings of WMT 2013*, pages 432–437, Sofia, Bulgaria, 2013.

David Kauchak and Regina Barzilay. Paraphrasing for automatic evaluation. In *Proceedings of HLT-NAACL 2006*, pages 455–462, New York City, 2006.

Michael Denkowski and Alon Lavie. Meteor Universal: Language specific translation evaluation for any target language. In *Proceedings of WMT 2014*, pages 376–380, Baltimore, Maryland, 2014.

Benjamin Marie, Atsushi Fujita, and Raphael Rubino. Scientific credibility of machine translation research: A meta-evaluation of 769 papers. In *Proceedings of ACL-IJCNLP 2021*, pages 7297–7306, Online, 2021.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. In *CoRR*, abs/1301.3781, 2013.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of ACL 2002*, pages 311–318, Philadelphia, Pennsylvania, 2002.

Ricardo Rei, Craig Stewart, Ana C. Farinha, and Alon Lavie. COMET: A neural framework for MT evaluation. In *Proceedings of EMNLP 2020*, pages 2685–2702, Online, 2020.

Thibault Sellam, Dipanjan Das, and Ankur Parikh. BLEURT: Learning robust metrics for text generation. In *Proceedings of ACL 2020*, pages 7881–7892, Online, 2020.

List of Publications

Publications Directly Related to the Thesis

1. Petra Barančíková and Ondřej Bojar. Intrinsic vs. Extrinsic Evaluation of Czech Sentence Embeddings: Semantic Relevance Doesn't Help with MT Evaluation. In *Proceedings of Machine Translation Summit XX: Volume 1*, pages 265–275, Geneva, Switzerland, August 2025. European Association for Machine Translation.
2. Jiří Bednař, Jakub Náplava, Petra Barančíková, and Ondřej Lisický. Some Like It Small: Czech Semantic Embedding Models for Industry Applications. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(21), pages 22734–22742, 2024.
3. Petra Barančíková and Ondřej Bojar. Costra 1.1: An Inquiry into Geometric Properties of Sentence Spaces. In *Proceedings of the 23rd International Conference on Text, Speech and Dialogue*, pages 135–143, Cham, Switzerland, September 2020. Springer.
4. Petra Barančíková and Ondřej Bojar. COSTRA 1.0: A Dataset of Complex Sentence Transformations. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3535–3541, Marseille, France, May 2020. European Language Resources Association.
5. Petra Barančíková and Ondřej Bojar. In Search for Linear Relations in Sentence Embedding Spaces. In *Proceedings of the 19th Conference ITAT 2019: Slovenskočeský NLP Workshop (SloNLP 2019)*, pages 125–132, Košice, Slovakia, September 2019. CreateSpace Independent Publishing Platform.
6. Aleš Tamchyna and Petra Barančíková. Manual and Automatic Paraphrases for MT Evaluation. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 3543–3548, Portorož, Slovenia, May 2016. European Language Resources Association.
7. Petra Barančíková and Rudolf Rosa. Targeted Paraphrasing on Deep Syntactic Layer for MT Evaluation. In *Proceedings of the Third International Conference on Dependency Linguistics (Depling 2015)*, pages 20–27, Uppsala, Sweden, August 2015. Uppsala University.
8. Petra Barančíková. Parmesan: Meteor without Paraphrases with Paraphrased References. In *Proceedings of the Ninth Workshop on Statistical Machine Translation*, pages 355–361, Baltimore, Maryland, USA, June 2014. Association for Computational Linguistics.
9. Petra Barančíková, Rudolf Rosa, and Aleš Tamchyna. Improving Evaluation of English–Czech MT through Paraphrasing. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 3402–3407, Reykjavík, Iceland, May 2014. European Language Resources Association.

10. Petra Barančíková and Aleš Tamchyna. Machine Translation within One Language as a Paraphrasing Technique. In *Proceedings of the 14th Conference ITAT 2014*, pages 1–6, Prague, Czechia, September 2014. Institute of Computer Science AS CR.
11. Petra Barančíková. Lexical Paraphrasing for Improvement of MT Evaluation. In *WDS'13 Proceedings of Contributed Papers*, pages 7–11, Prague, Czechia, 2013. Matfyzpress.

Other Publications

1. Václava Kettnerová, Markéta Lopatková, Anna Vernerová, and Petra Barančíková. Towards a Semi-Automatic Detection of Reflexive and Reciprocal Constructions and Their Representation in a Valency Lexicon. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3136–3144, Marseille, France, May 2020. European Language Resources Association.
2. Václava Kettnerová, Markéta Lopatková, Eduard Bejček, and Petra Barančíková. Enriching VALLEX with Light Verbs: From Theory to Data and Back Again. *The Prague Bulletin of Mathematical Linguistics, 111*, pages 29–56, 2018.
3. Petra Barančíková and Václava Kettnerová. Paraphrases of Verbal Multiword Expressions: The Case of Czech Light Verbs and Idioms. In *Multiword Expressions at Length and in Depth: Extended Papers from the MWE 2017 Workshop*, pages 35–59, Berlin, Germany, 2018. Language Science Press.
4. Petra Barančíková and Václava Kettnerová. ParaDi: Dictionary of Paraphrases of Czech Complex Predicates with Light Verbs. In *Proceedings of the 13th Workshop on Multiword Expressions (MWE 2017)*, pages 1–10, Valencia, Spain, April 2017. Association for Computational Linguistics.
5. Václava Kettnerová, Petra Barančíková, and Markéta Lopatková. Lexicographic Description of Complex Predicates in Czech: Between Lexicon and Grammar. In *Proceedings of the XVII EURALEX International Congress: Lexicography and Linguistic Diversity*, pages 893–904, Tbilisi, Georgia, September 2016. Tbilisi University Press.